

TE Curve Verification Pipeline Familywise ONNX Report - residual_harmonic_mlp

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/residual_harmonic_mlp/2026-07-07-15-28-55__track2_residual_harmonic_mlp_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/residual_harmonic_mlp/2026-07-07-15-28-55__track2_residual_harmonic_mlp_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_mlp/2026-07-07-15-28-55__track2_residual_harmonic_mlp_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_mlp/2026-07-07-15-28-55__track2_residual_harmonic_mlp_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `residual_harmonic_mlp` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_residual_harmonic_mlp_fw_simplified_setpoints</code>	<code>2026-07-07-10-54-26_te_residual_harmonic_mlp_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_residual_harmonic_mlp_bw_simplified_setpoints</code>	<code>2026-07-07-11-09-49_te_residual_harmonic_mlp_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_residual_harmonic_mlp_global_simplified_setpoints</code>	<code>2026-07-07-10-46-08_te_residual_harmonic_mlp_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/forward/2026-07-07-10-54-26_te_residual_harmonic_mlp_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/forward/2026-07-07-10-54-26_te_residual_harmonic_mlp_fw_simplified_setpoints/python/residual_harmonic_mlp-epoch=079-val_mae=0.00306417.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/backward/2026-07-07-11-09-49_te_residual_harmonic_mlp_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/backward/2026-07-07-11-09-49_te_residual_harmonic_mlp_bw_simplified_setpoints/python/residual_harmonic_mlp-epoch=032-val_mae=0.00306476.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/global/2026-07-07-10-46-08_te_residual_harmonic_mlp_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_mlp/global/2026-07-07-10-46-08_te_residual_harmonic_mlp_global_simplified_setpoints/python/residual_harmonic_mlp-epoch=010-val_mae=0.00315844.ckpt</code>

Aggregate Metrics

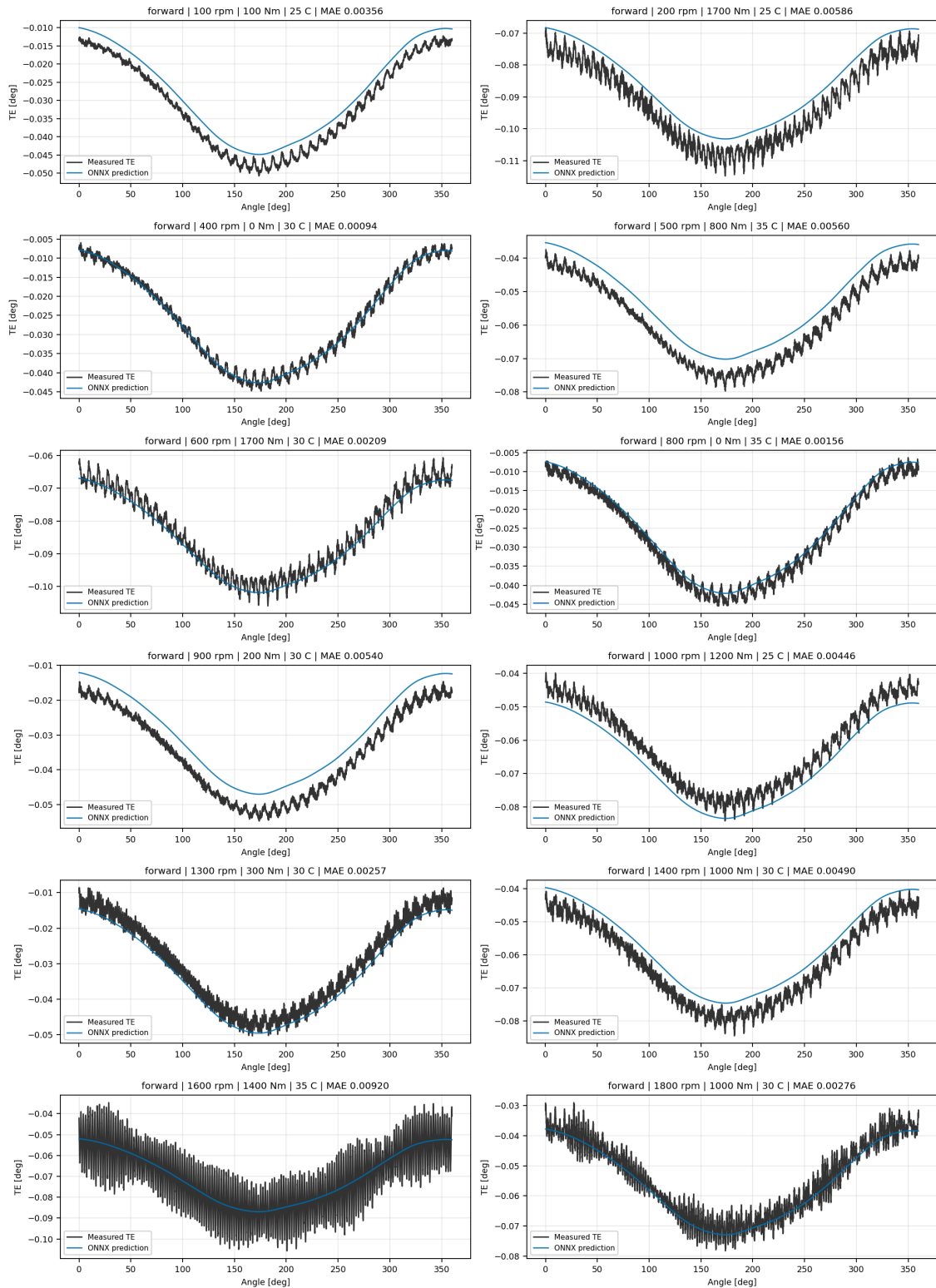
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003329	0.003739	7.380	13.427
backward	97	0.003734	0.004148	8.181	15.080
global	194	0.003728	0.004144	8.241	15.361

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000903	0.002727	0.001697	0.009765
backward	0.001054	0.003027	0.001828	0.009713
global	0.001222	0.003092	0.001781	0.009788

Forward 12-Curve Page

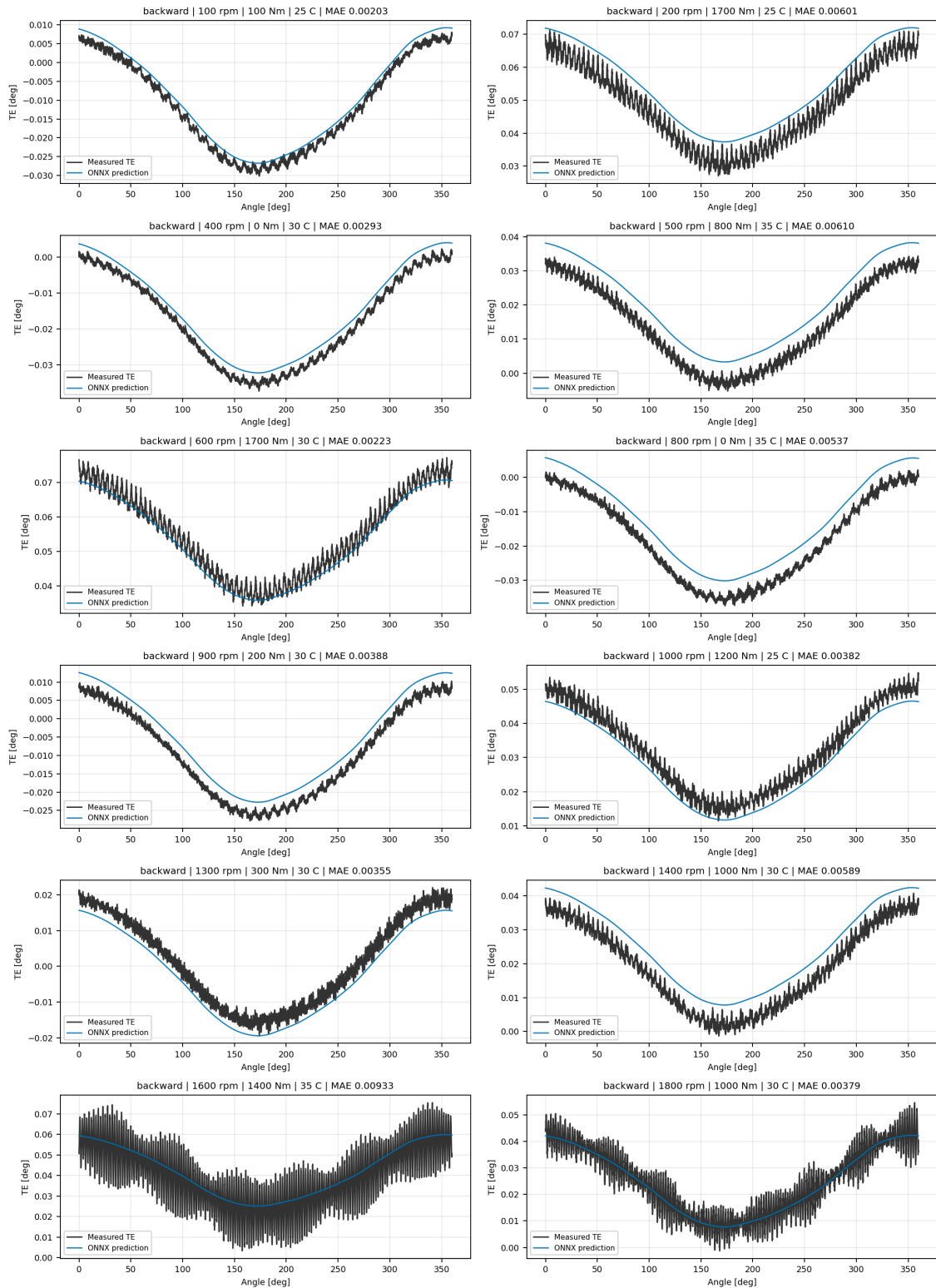
Simplified Dataset + Setpoints | residual_harmonic_mlp | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

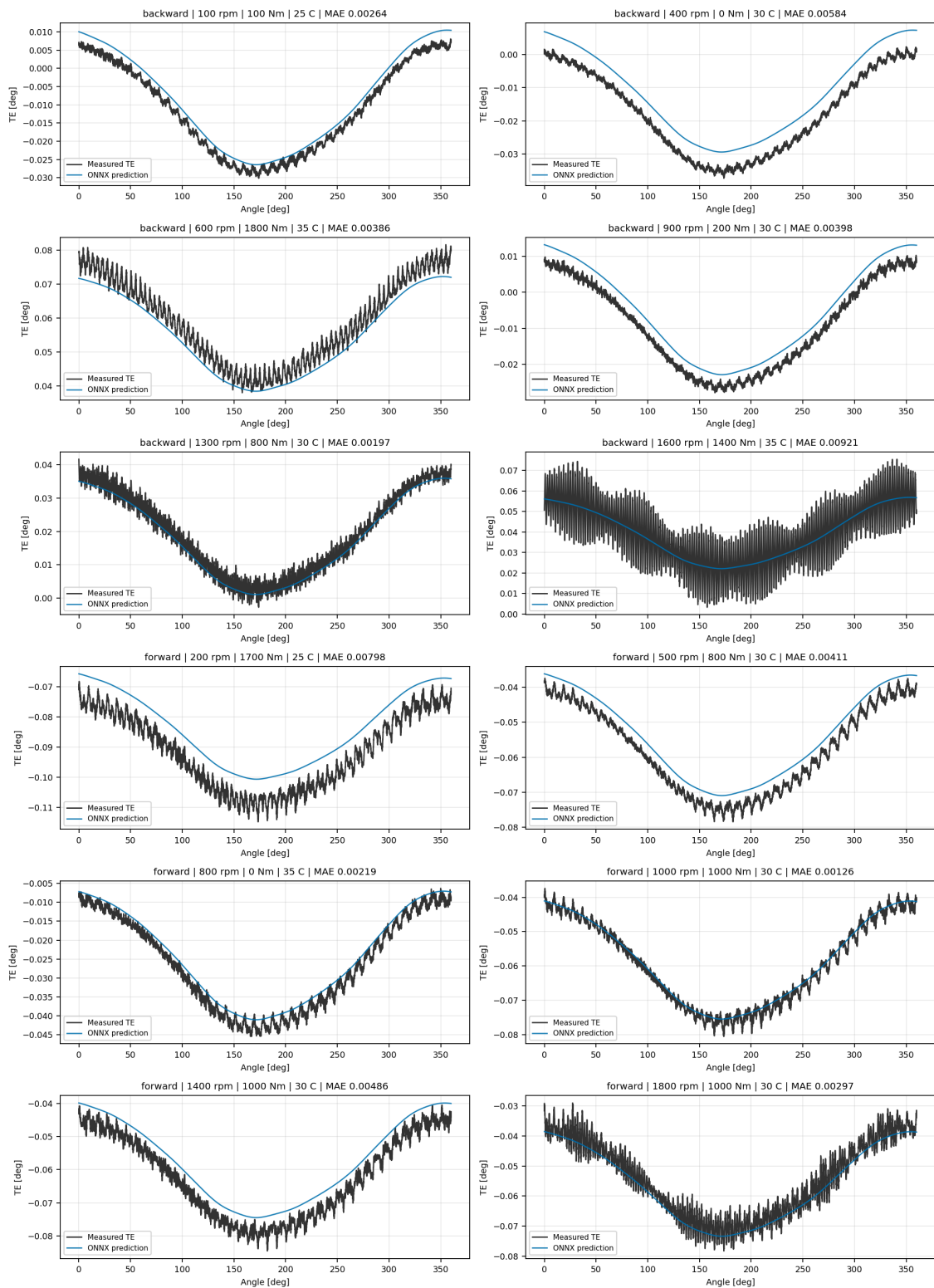
Simplified Dataset + Setpoints | residual_harmonic_mlp | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | residual_harmonic_mlp | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: residual_harmonic_mlp ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_mlp_fw_polished_setpoints	2026-07-07-12-02-44_te_residual_harmonic_mlp_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_residual_harmonic_mlp_bw_polished_setpoints	2026-07-07-12-36-55_te_residual_harmonic_mlp_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_residual_harmonic_mlp_global_polished_setpoints	2026-07-07-11-34-39_te_residual_harmonic_mlp_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/forward/2026-07-07-12-02-44_te_residual_harmonic_mlp_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/forward/2026-07-07-12-02-44_te_residual_harmonic_mlp_fw_polished_setpoints/python/residual_harmonic_mlp-epoch=122-val_mae=0.00159866.ckpt
backward	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/backward/2026-07-07-12-36-55_te_residual_harmonic_mlp_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/backward/2026-07-07-12-36-55_te_residual_harmonic_mlp_bw_polished_setpoints/python/residual_harmonic_mlp-epoch=093-val_mae=0.00162645.ckpt
global	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/global/2026-07-07-11-34-39_te_residual_harmonic_mlp_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_mlp/global/2026-07-07-11-34-39_te_residual_harmonic_mlp_global_polished_setpoints/python/residual_harmonic_mlp-epoch=069-val_mae=0.00158201.ckpt

Aggregate Metrics

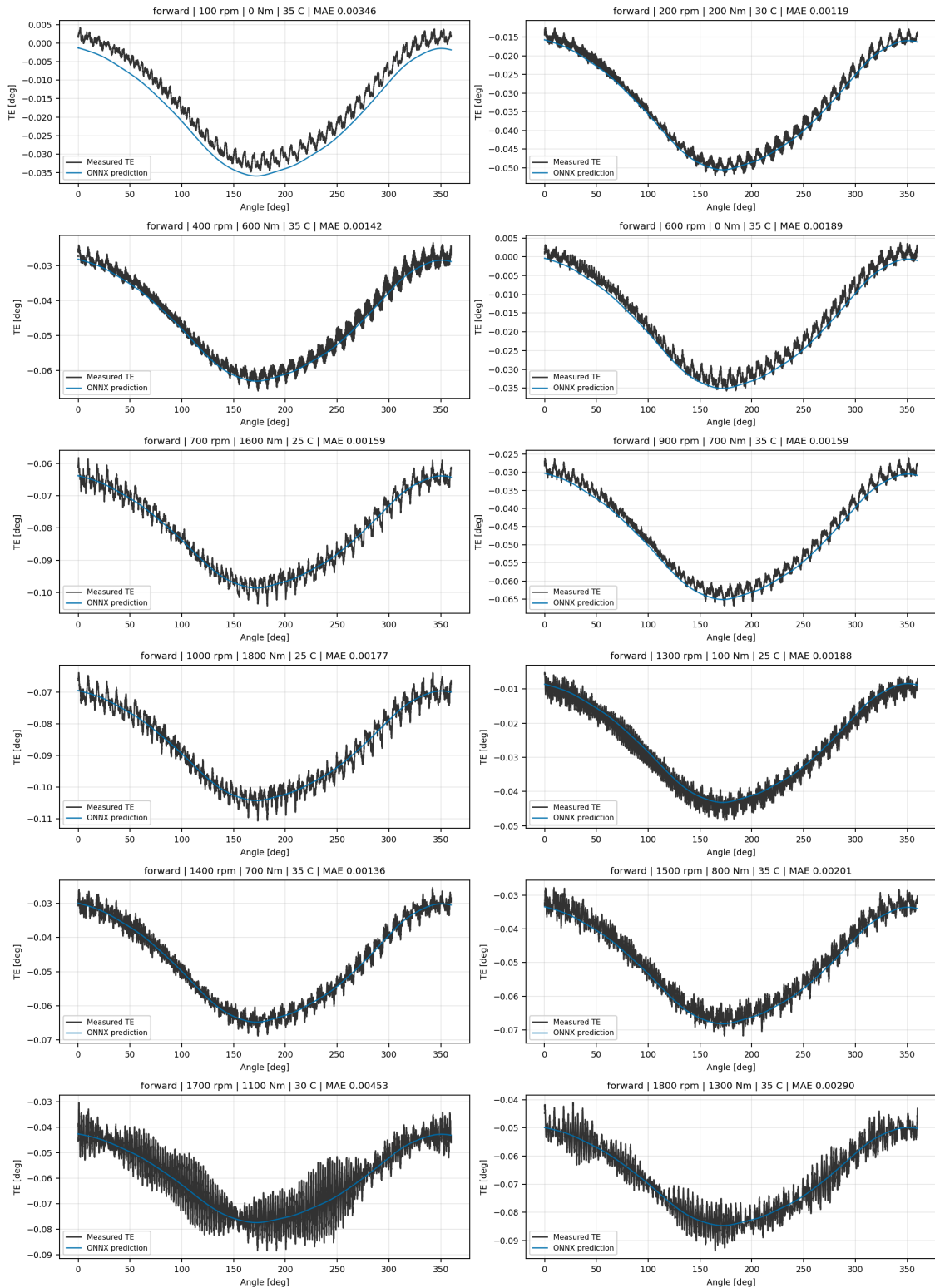
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002201	0.002648	4.569	8.932
backward	94	0.002763	0.003257	4.912	10.607
global	194	0.002465	0.002939	4.704	10.418

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000413	0.000863	0.001928	0.010958
backward	0.000710	0.001022	0.002359	0.012106
global	0.000172	0.000937	0.002139	0.011590

Forward 12-Curve Page

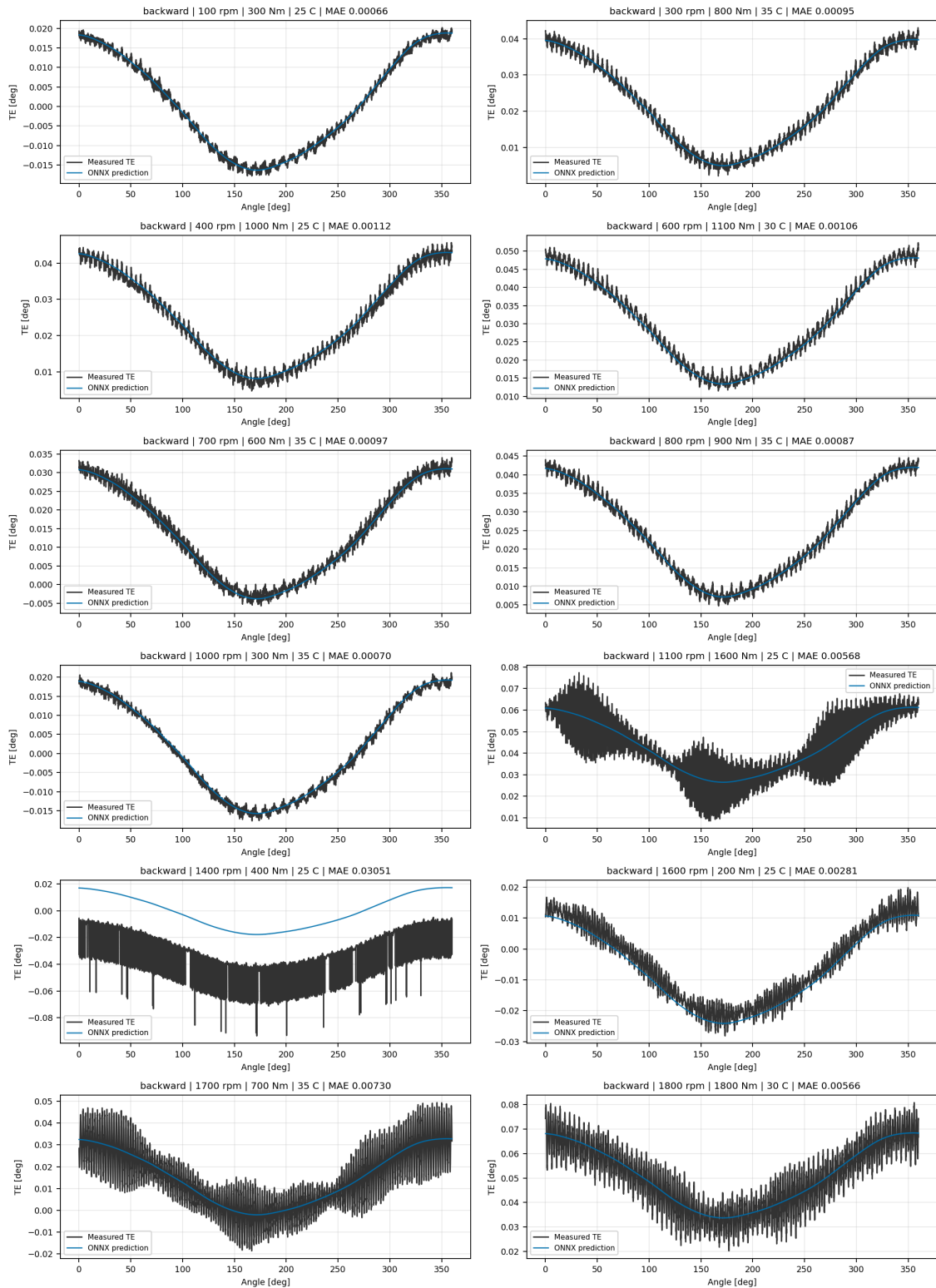
Polished Dataset + Setpoints | residual_harmonic_mlp | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

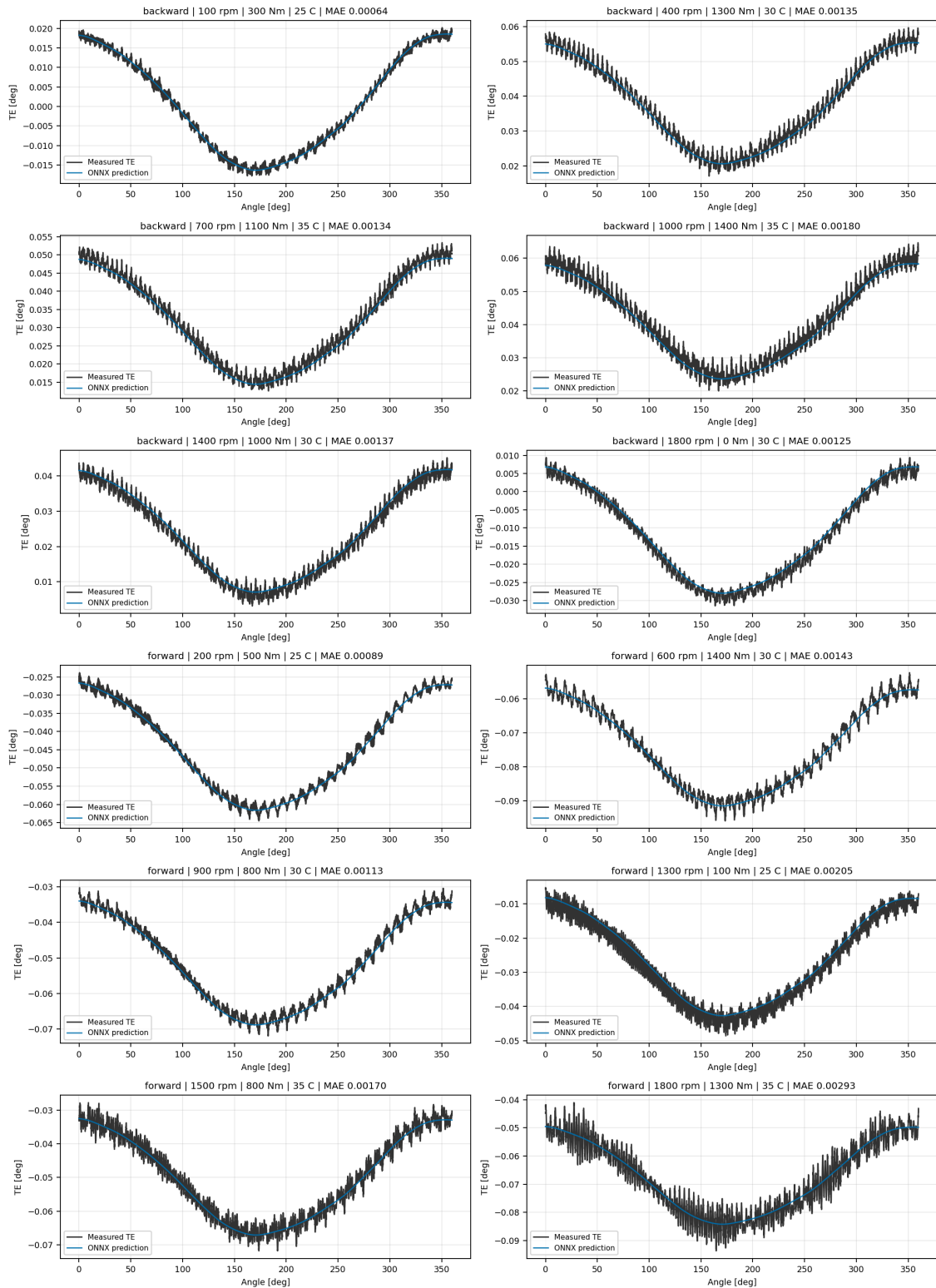
Polished Dataset + Setpoints | residual_harmonic_mlp | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | residual_harmonic_mlp | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: residual_harmonic_mlp ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_mlp_fw_polished_actual_values	2026-07-07-13-50-19_te_residual_harmonic_mlp_fw_polished_actual_values	polished_point_v1
backward	te_residual_harmonic_mlp_bw_polished_actual_values	2026-07-07-14-15-24_te_residual_harmonic_mlp_bw_polished_actual_values	polished_point_v1
global	te_residual_harmonic_mlp_global_polished_actual_values	2026-07-07-13-20-06_te_residual_harmonic_mlp_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/forward/2026-07-07-13-50-19_te_residual_harmonic_mlp_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/forward/2026-07-07-13-50-19_te_residual_harmonic_mlp_fw_polished_actual_values/python/residual_harmonic_mlp-epoch=052-val_mae=0.00163893.ckpt
backward	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/backward/2026-07-07-14-15-24_te_residual_harmonic_mlp_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/backward/2026-07-07-14-15-24_te_residual_harmonic_mlp_bw_polished_actual_values/python/residual_harmonic_mlp-epoch=075-val_mae=0.00160615.ckpt
global	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/global/2026-07-07-13-20-06_te_residual_harmonic_mlp_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_mlp/global/2026-07-07-13-20-06_te_residual_harmonic_mlp_global_polished_actual_values/python/residual_harmonic_mlp-epoch=077-val_mae=0.00160259.ckpt

Aggregate Metrics

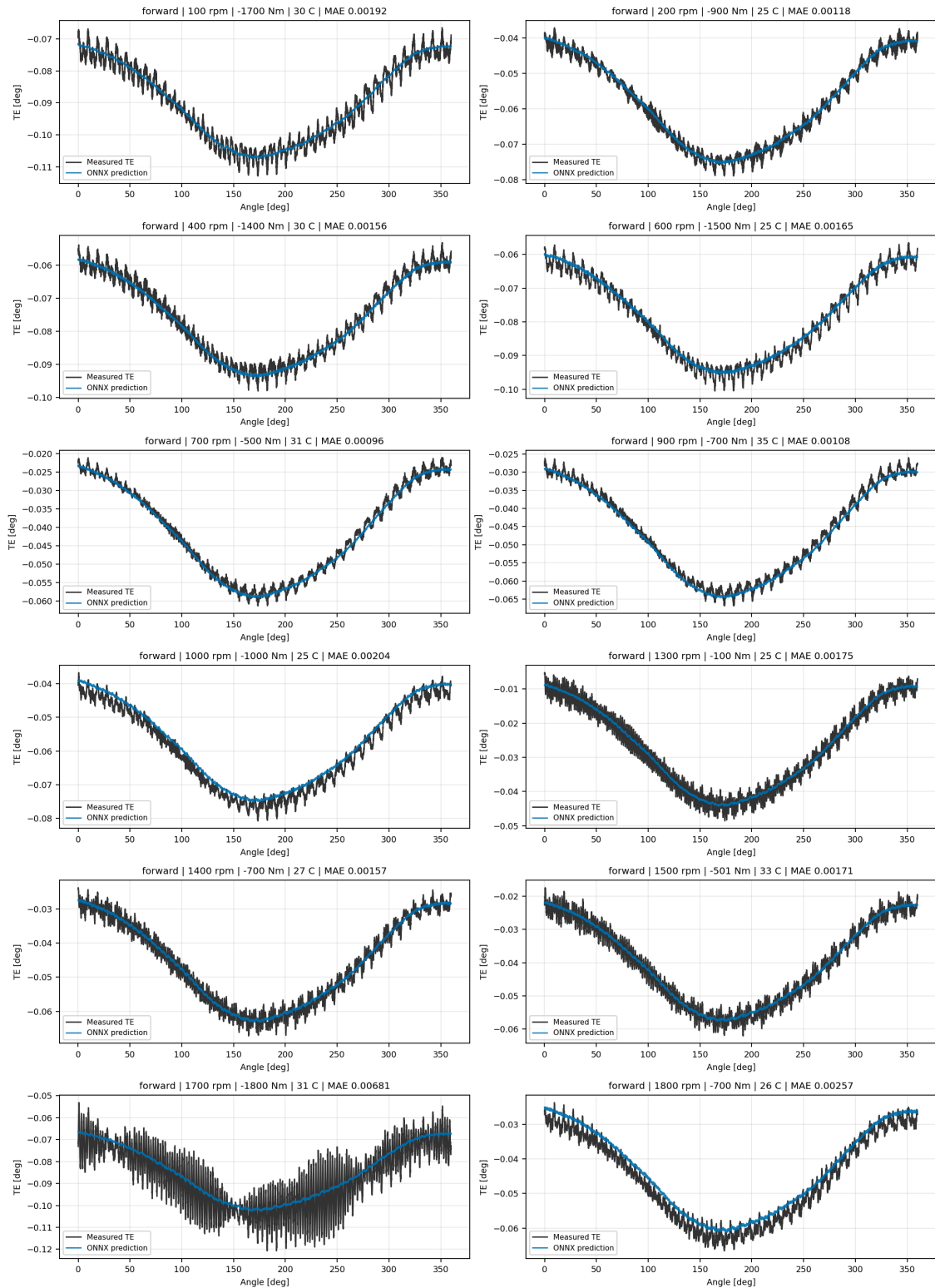
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002191	0.002649	4.524	8.799
backward	94	0.002770	0.003275	4.914	10.607
global	194	0.002434	0.002915	4.627	9.999

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000077	0.000859	0.001949	0.009716
backward	0.000674	0.000970	0.002409	0.010930
global	0.000291	0.000869	0.002171	0.010348

Forward 12-Curve Page

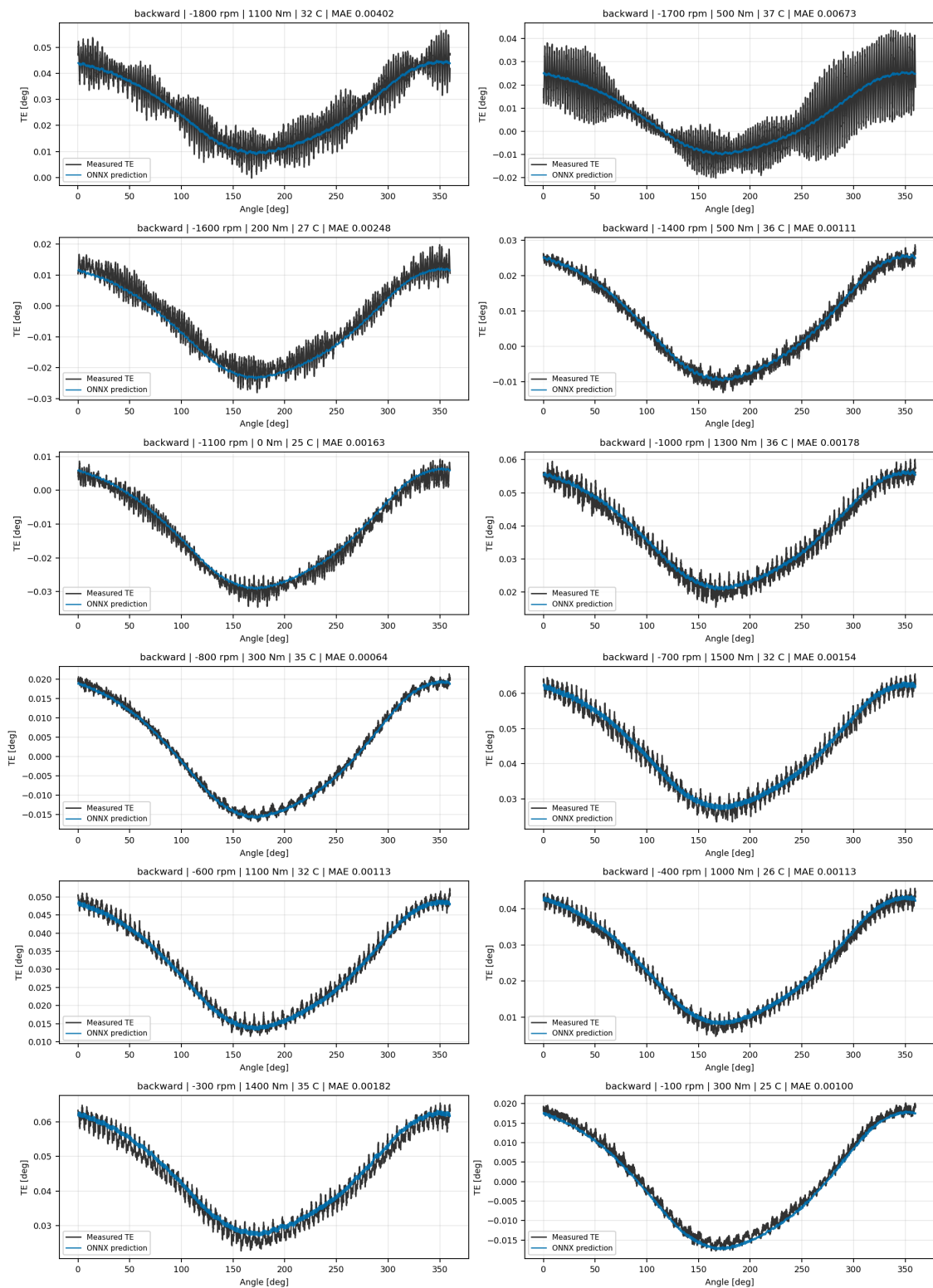
Polished Dataset + Actual Values | residual_harmonic_mlp | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

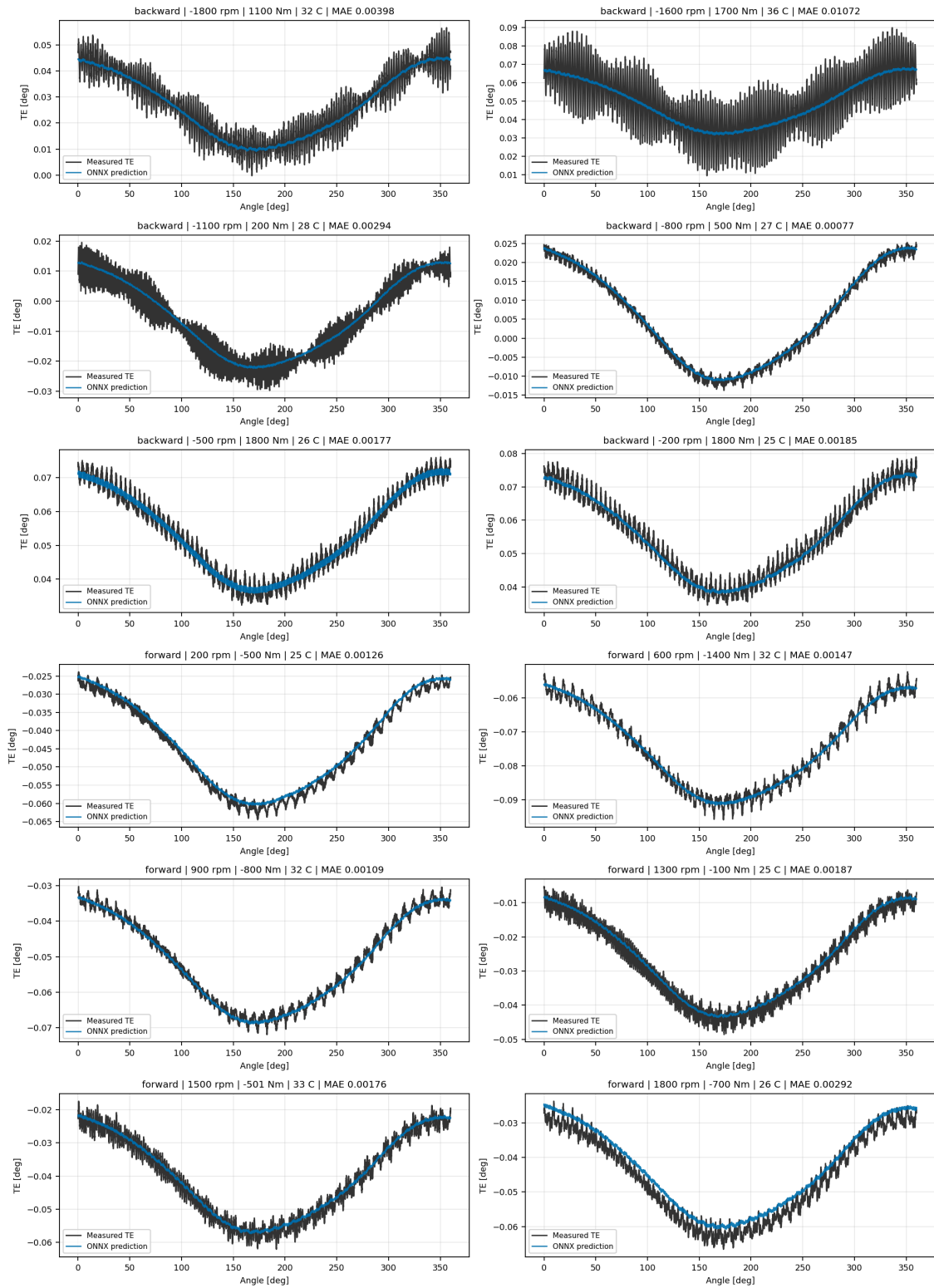
Polished Dataset + Actual Values | residual_harmonic_mlp | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | residual_harmonic_mlp | global



polished_dataset_actual_values global 12-curve collage